

Programme	B. Sc. Computer Science				
Course Title	INTRODUCTION TO IT				
Type of Course	Minor				
Semester	I				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic understanding of computer operation Basic Science fundamentals				
Course Summary					
The course will create an overall generic awareness about scope of the field of IT and to impart basic personal computing skills and will create background knowledge for the various courses in the programme.					

Course Outcomes (CO): .

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand basic terminology in the field of IT	U	C	Instructor-created exams / Assignment
CO2	Identify and describe essential computer hardware components.	U	C	Viva Voce
CO3	Comprehend the distinction between system software and application software and their respective roles in computer functionality	U	C	Practical / Group Work
CO4	Produce documents with precision and efficiency using LaTeX	Ap	P	Practical / Group Work

CO5	Understand the basics of networking and internet concepts.	U	C	Practical/Exam/ Assignments
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Characteristics of Computers		10	15
	1	Characteristics: Automatic, speed, accuracy, memory, diligence; Digital signals, Instruction set	2	
	2	Evolution of computers and generation of computers	2	
	3	Classification of computers: Microcomputer, Minicomputer, mainframes, Supercomputers; Personal computers: Desktop,Laptops	2	
	4	Binary System and data representation (BCD,ASCII,Unicode)	3	
	5	Hardware & Software,Von Neumann model.	1	
II	Hardware		13	19
	6	CPU- CU, ALU, Registers	2	
	7	Memory units: RAM(SDRAM, DRAM)- feature wise comparison only); ROM-(PROM,EPROM,EEPROM)	2	
	8	Auxiliary storage: Flash memory ,Magnetic devices, HDD SSD,	2	
	9	Input devices - keyboard, mouse, scanner, speech input devices, digital camera, Touch screen, Joystick, Optical readers, bar code reader	3	
	10	Output devices: Display device-Mouse, Joystick, Touchscreens, LCD, Plasma, LED, Printers: Dot-matrix, Inkjet, Laser; Plotters, speaker.	4	
III	Software		12	20
	11	System software, Application software ,examples	1	

	12	Operating systems: Single user, Multitasking, Time-sharing ,multi-user;	1	
	13	Basic features of OS: Process management, Memory management, Device Management,	2	
	14	Booting, POST	1	
	15	Computer Viruses & Protection	2	
	16	Free software, Open source	1	
	17	LaTeX : Introduction, installation, and basic document creation,Text styling, sectioning, and lists, Citations and references, Inserting images and creating tables.	4	
IV	Computer Networks		10	16
	18	Requirements for a network	1	
	19	Server, Workstation, switch, router, network operating systems	2	
	20	Internet: brief history, World Wide ,Web, Websites, URL, Browsers, Search engines	2	
	21	Internet connections: ISP, Dial-up, cable modem	2	
	22	Characteristics of web-based systems, Web pages, introduction to HTML.	3	
V	Practical Applications, Case Study and Course Project		30	
	1	1. Document Basics: Create a document with a title, author, and date. 2. Sections and Headings: Add sections and subsections with headings. 3. Lists: Insert bulleted and numbered lists. 4. Graphics: Insert images and adjust their placement. 5. Tables: Create tables with rows and columns. 6. References: Add citations and create a bibliography. 7. Formatting: Apply styles like bold, italics, and underline.	20	

	2	Case study: 1. Academic Essay: Write an essay on a chosen topic, formatting headings, paragraphs, and citations using LaTeX commands. 2. Research Paper: Write a structured paper on a chosen topic, incorporating sections, citations, and formatting using LaTeX.	10	

References

1. P. K Sinha, Fundamentals of Computers
2. Behrouz A Forouzan, Data Communication & Networking, MC Graw Hill

Reference Books:

3. Tanenbaum, Andrew S. and Herbert Bos. Modern Operating Systems. 4th ed., Pearson, 2014.
4. Norton, Peter. Introduction to Computers. 7th ed., McGraw-Hill, 2016.
5. Harel, David. Algorithmics: The Spirit of Computing. 3rd ed., Addison-Wesley, 2004.
6. LaTeX Beginner's Guide - Second Edition. Author(s): Stefan Kottwitz. Publisher(s): Packt Publishing.

Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	1	1	-	-	-	-						

CO 2	2	2	-	-	-	-						
CO 3	2	2	-	-	1	-						
CO 4	-	-	-	-	2	-						
CO 5	-	-	-	-	1	-						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓		✓